



CRYSTAL CALIBRATION SALES AND SERVICE CO., LTD.

45/48 Soi Salathammasop31, Salathammasop Rd.,

Salathammasop, Thawewatthana, Bangkok 10170 Thailand

Tel : 0-2408-8474-5 Fax : 0-2408-8477 Email : info@crystalcal.com www.crystalcal.com



CERTIFICATE OF CALIBRATION

Certificate No. : 25-1390-136

Issue Date : 5 November 2025

Work Order No. : 25/1390

Customer Name : กลุ่มงานเทคนิคการแพทย์ โรงพยาบาลพระจอมเกล้า จังหวัดเพชรบุรี
53 ถนนไฟ ต.คลองกระแซง
อ.เมือง จ.เพชรบุรี 76000

Date of Received : 28 October 2025

Date of Calibration : 28 October 2025

Instrument Details : Description : Temperature Controlled Enclosures [Freezer]
Manufacturer : ANTECH
Model : MDF-40U525
Serial No. : 5452 5070 0009 272G 0007
ID No. : 6530-032-0003-1/63
Resolution : 0.1 °C
Location : Special Lab

Calibration Method : This instrument was calibrated by insert standard thermometer into the chamber according to calibration procedure no. CWI-T-10 follow up to TLAS G-20-1/02-08
(E) : Guidelines for Calibration and Checks of Temperature Controlled Enclosures.

Environmental Conditions :

Temperature : Area Monitoring between 15°C to 40°C

Humidity : Area Monitoring between 30%RH to 85%RH

Line Voltage : Area Monitoring 220 VAC $\pm$ 10%

Traceability of Measurement :

This certificate of calibration documents the traceability to national standard, which realize the unit of measurement according to the International system of Units (SI) and The temperature scale in use at this laboratory is The International Temperature scale of 1990.

Calibrated by : Mr. Kritsada Kaewwangpa
Calibration Engineer

Approved by : 
(Mr. Anuwat Yaklermjit)
Laboratory Manager

This certificate may not be reproduced other than in full except with the prior written approval of Crystal Calibration Sales and Service co., Ltd.

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Details of Calibration

1. Reference Standards Instrument

Instrument	Model	Serial No./Ins No.	Certificate No.	Due Date
Data Acquisition unit	34972A	MY59002085	25-0861-027	11 July 2026
Sensor type	TC Type T	CH # 301-309	25-0861-027	11 July 2026

2. Certificate traceable

: This certificate traceable to The International System of Unit refer to
Crystal Calibration Sales and Service Co., Ltd. , NAC Calibration No. 0260

3. Condition of item

: Used

4. Calibration site

: On - Site

5. Result of Calibration

: Without adjustment

6. Evaluate Condition

: Time Constant : 3 Hour 19 Minute At cal. point -40 °C
Air vent : Off
Fan speed status : None Fan Speed

7. Calibration note

: The results reported in this certificate refer to the condition of instrument on
the process into the steady state of chamber

8. Sensors Installation Diagram

: When ; Sensor installation location in Chamber @ Working Space

A = Distance between sensor and wall of chamber is 10 cm

9. Dimensions of chamber

: W = 0.69 m ; D = 0.55 m ; H = 1.16 m

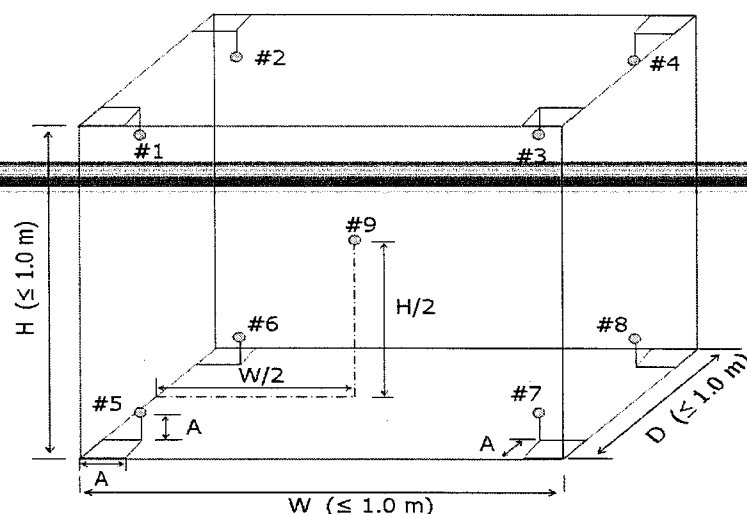


Diagram of Chamber

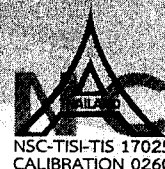


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Result of Temperature Distribution and Performance Check

Table1 : Reporting of Temperature Distribution

Calibration point (°C)	Average Measured Temperature (°C) @ Sensor No. (Sensor No.9 is REF)									Uncertainty ± (°C)
	#1	#2	#3	#4	#5	#6	#7	#8	#9	
-40	-38.8	-38.5	-38.6	-38.7	-36.6	-36.6	-36.2	-35.5	-37.9	2.6

Table 2 : Reporting of Performance check

Calibration Point (°C)	Indicator Set Point (°C)	Indicator Reading (°C)			Stability ± (°C)	Uniformity (°C)	Overall variation (°C)
		MAX	MIN	Average			
-40	-42	-39	-41	-40	0.2	2.6	3.8

Note

Customer can't removed items form chamber and this report customer request and accepted in certificate

The reference sensor is preferably located of the geometric center of chamber

The measured temperature data readout by software "Benchlink Datalogger 4"

The quoted uncertainty include " Stability " and " Loading effect (20% of Temp Uniformity) "

Stability - one-half of the greatest maximum difference of measured temperatures at any one sensor.

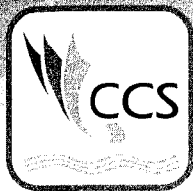
Uniformity - the maximum difference of measured temperatures at any sensors and the measured temperature at the reference location which are observed at the same time or at as close an observation time as possible to determine the temperature pattern or homogeneity within the chamber under steady state conditions.

Overall Variation - The difference of the maximum and minimum measured temperatures throughtout observation time.

Indicating Temperature - the average reading of indicating device that forms the integral part of the enclosure.

This result of calibration was found accurate as shown on date and place of calibration only.

The reported uncertainty of measurement was based on a standard uncertainty multiplied by a coverage factor $k=2$ providing a level of confidence of approximately 95%.



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Statements of conformity report

Refer Certificate No. : 25-1390-136

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Work Order No. : 25/1390

Detail of Equipment

Description : Temperature Controlled Enclosures [Freezer]

Manufacturer : ANTECH

ID No. : 6530-032-0003-1/63

Model : MDF-40U525

Serial No. : 5452 5070 0009 272G 0007

Result of Calibration

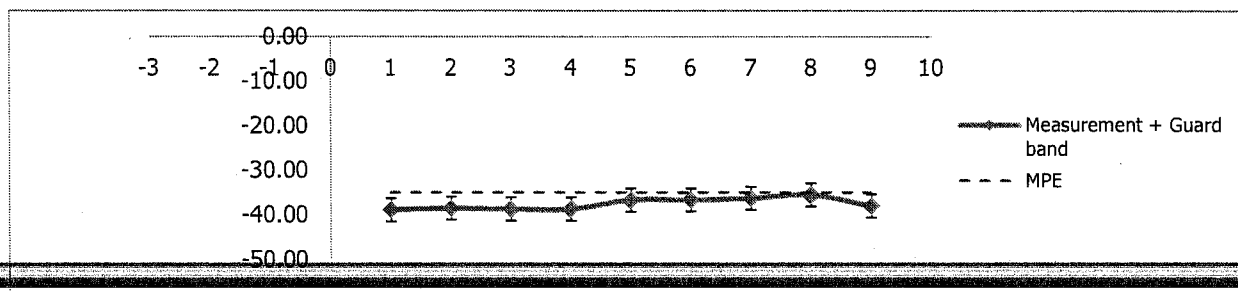
This result of calibration : Without adjustment

Resolution : 0.1 °C

Calibration point	-40 °C
MPE below	-35 °C

Uncertainty		Uncertainty MPE		Evaluate resulte
2.6	°C	3.0	°C	Passed

Position	Standard reading	Measurement decide	MPE below	Evaluate resulte
#1	-38.8	-38.8	-35	Passed
#2	-38.5	-38.5	-35	Passed
#3	-38.6	-38.6	-35	Passed
#4	-38.7	-38.7	-35	Passed
#5	-36.6	-36.6	-35	Passed
#6	-36.6	-36.6	-35	Passed
#7	-36.2	-36.2	-35	Passed
#8	-35.5	-35.5	-35	Passed
#9	-37.9	-37.9	-35	Passed



This statement judgment and decision by Laboratory guideline only

The conformity certificate documents validity following ISO/IEC Guide 98-4 : Role of measurement uncertainty in conformity assessment based on statement with measurement refer to ISO 7218 : 2024 ; 6.3.6.1 consider expanded measurement uncertainty (k=95%)

The tolerance and decision rules ;

Decision rules : Laboratory decide temperature measurement in chamber is ;

MPE of Customer or Lab decide = Measurement any points ; $\leq$ MPE is pass , $>$ MPE is Fail

Statements of conformity decision by :

(Mr. Anuwat Yaklermjit)

Laboratory Manager